MIL-P-83668A
8 April 1988
SUPERSEDING
MIL-P-83668 (USAF)
26 August 1976

MILITARY SPECIFICATION

PLASTIC BOARD (FOR PACKAGING APPLICATIONS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plastic board primarily for use in the fabrication of shipping containers. Plastic board may also be used in the fabrication of pads, sleeves, box liners, partitions and cut shapes.

1.2 Classification. Plastic boards shall be of the following types:

Type I - Profile extrusion
Type II - Laminated corrugated

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Air Force Packaging Evaluation Activity, HQ AFLC/DSTZT, Wright-Patterson AFB, Ohio 45433-5999 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8135

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MILITARY SPECIFICATIONS:

MIL-P-116 - Preservation, Method of

MILITARY STANDARDS:

MIL-STD-129 - Marking for Shipment and Storage

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

21 CFR - Federal Food, Drug, and Cosmetic Act, Food Additive Amendment:

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P. Street, N. W., Washington DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS:

D256 - Impact Resistance of Plastics and Electrical Insulating Materials, Test Methods For

D646 - Grammage of Paper and Paperboard (Weight per unit area).

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D647 - Design of Molds for Test Specimens of Plastic Molding Materials, Practice For

D648 - Deflection Temperature of Plastics Under Flexural Load, Test Method For

(Application for copies of ASTM standards by serial designation and title should be made to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103).

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI) STANDARDS:

T411 - Thickness (caliper) of Paper, Paperboard, and Combined Board.

(Applications for copies of TAPPI standards by serial designation and title should be made to Technical Association of the Pulp and Paper Industry, PO Box 105113, Atlanta GA 30348).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials. The plastic board shall be produced from a single material of uniform composition. The material shall be polyethylene, polypropylene or ethylene or propylene copolymers. When the plastic board is available in either polyethylene, polypropylene or copolymers, the procuring activity may specify the desired material. When specified that the plastic board is intended for food applications, the material shall conform to the Federal Food, Drug, and Cosmetic Act, Food Additive Amendment, Paragraph 121.2501 of Subpart F (see 4.4.3 and 6.2).

3.3 Color. As specified by procuring activity (see 4.4.3 and 6.2).

3.4 Deflection temperature at 66 psi. The material from which the board is produced shall have a minimum deflection temperature of 150 degrees F at a fiber stress of 66 psi when tested as specified in 4.5.2.

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3.5 Notched Izod impact strength at 0 degrees F. The material from which the board is produced shall have a minimum notched Izod impact strength of 0.7 ft-lb/in at 0 degrees F when tested as specified in 4.5.3.

3.6 Form. The finished plastic board shall be furnished in flat sheets. The sheets shall lay flat within 1/8" when tested in accordance with 4.5.8.

3.7 Construction.

3.7.1 Type I. Type I plastic board shall be a profile extrusion with two facings separated by ribs (see 4.4.3).

3.7.2 Type II. Type II plastic board shall be in a laminated structure with two facings separated by a corrugating medium (see 4.4.3).

3.8 Physical properties.

3.8.1 Basis weight. The basis weight shall not be less than the lower limit nor greater by more than 5 percent of the upper limit of the basis weight range specified (see 6.2) when tested as specified in 4.5.4.

3.8.2 Thickness. The thickness shall be that provided by the manufacturer for the basis weight specified. No single thickness measurement shall vary by more than ± 10 percent from the manufacturer's designated thickness when tested as specified in 4.5.5.

3.9 Bending qualities. The plastic board shall show no visual surface breaks or cracks when tested as specified in 4.5.6.

3.10 Flute T-peel strength (type II only). Type II plastic board, with basis weight of 140 pounds per 1000 square feet or less, shall have a minimum flute T-peel strength of 5 pounds. Plastic board, with basis weight greater than 140 pounds per 1000 square feet, shall have a minimum flute T-peel strength of 35 pounds. Flute T-peel strength shall be determined as specified in 4.5.7.

3.11 Dimensions. Sheet size, length and width, shall be as specified by the procuring activity (see 6.2).

3.11.1 Dimensional tolerances. The dimensional tolerances shall be minus 0, + 1/2 inch for both length and width of the specified sheet size (see 4.4.3).

3.12 Workmanship. The plastic board shall contain no foreign matter except dust or dirt that adheres due to the tendency of plastics to accumulate electric charges. There shall be no scuffs, blisters, tears or checks. Trimmed edges must be perpendicular and free from burrs. Dirt or dust particles, which adhere to the plastic board because of the tendency of plastics to accumulate electric charges, shall be disregarded in the examination for cleanliness (see 4.4.3).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of examinations and tests for all the requirements of this specification. Approval of the first article inspection does not preclude the requirements for performing the quality conformance inspection. First article inspection may be waived when the procuring activity or contract administration has data or other evidence to indicate that prior successful first article inspection has been conducted (see 6.2). First article inspection shall be conducted whenever a change occurs in the manufacturing process or materials used. A wavier should be considered when a contractor is in continuous production between contracts (see 3.1).

4.3.1 First article samples. First article samples shall be produced using the same production processes, procedures, equipment and material used in fulfilling the contract. First article samples shall be examined and tested as specified in 4.3.2 and shall consist of a minimum of two sheets, of the size specified in the purchase order, with a minimum total area of 36 square feet. Failure of any test specimen from the first article samples to meet the specified requirement shall be cause for rejection (see. 3.1).

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4.3.2 First article examination and testing. First article sample sheets shall be examined for defects listed in 4.4.3 and shall then be subjected to the applicable tests listed in Table I. The material used in producing the test specimens for the deflection temperature at 66 psi and the notched Izod impact tests shall be from the same lot of material used in fulfilling the contract (see 3.1).

4.4 Quality Conformance. Quality conformance shall consist of the examinations listed in 4.4.3 and 4.4.4.

TABLE I

First Article Testing

Property	Requirement Para	Test Method Para	No. of Determinations
Deflection temperature at 66 psi	3.4	4.5.2	5
Notched Izod impact strength at 0 degrees F	3.5	4.5.3	3
Basis weight	3.8.1	4.5.4	2 1/
Thickness	3.8.2	4.5.5	10 2/
Bending qualities	3.9	4.5.6	2 3/
Flute T-peel strength (type II only)	3.10	4.5.7	10 4/

NOTES:

1/ The basis weight shall be determined for each of two sheets.

2/ The thickness of five test specimens from each of two sheets shall be determined.

3/ The bending qualities of one test specimen from each of two sheets shall be determined.

4/ The flute T-peel strength of five test specimens from each of two sheets shall be determined.

4.4.1 Inspection lot. A quality conformance inspection lot shall consist of the ordered quantity. The lot size, for determining sample

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size, shall be expressed in units of the end item (number of sheets for examination under 4.4.3 and in units of unit pack under 4.4.4).

4.4.2 Sample size and acceptable quality levels. Samples, selected at random, shall be examined for the defects listed in 4.4.3 and 4.4.4. The sample sizes are given in 4.4.2.1 and 4.4.2.2.

4.4.2.1 Lot sizes of 1 to 1200.

Lot Size	Sample Size
1	1
2 to 25	2
26 to 150	3
151 to 1200	5

4.4.2.2 Lot sizes greater than 1200.

Lot Size	Sample Size N_1	Sample Size N_2
1201-35000	8	16
over 35000	13	26

4.4.3 Visual and dimensional inspection. The end item shall be examined for the defects listed in the following table. No more than two sample units shall be selected from any one shipping unit.

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TABLE II

CLASSIFICATION OF DEFECTS

Examine	Defect	Classification
Workmanship	Surfaces not clean, except that dirt or dust adhering because of electric charge accumulation shall be disregarded. Tear, hole, cut, scuff, marks, blisters, splices (see 3.12).	Minor
Length and width	Less than specified or more than plus 1/2 inch of specified size (see 3.11, 3.11.1 and 6.2).	Major
Material	Not as specified (see 3.2 and 6.2).	Major
Construction	Not as specified (see 3.7 and 6.2).	Major
Color	Not as specified (see 3.3 and 6.2).	Major
Markings (shipping interior packages (see 5.3)	Incorrect, illegible, improper location.	Minor

4.4.4 Inspection of packaging. The unit packs shall be examined for defects listed in Table II. Unit packs may be examined prior to closure (see 4.4.3).

4.5 Test methods.

4.5.1 Conditioning. Flatness, basis weight, thickness and bending qualities test specimens shall be conditioned for a minimum of 24 hours at 65 to 85 degrees F. Heat deflection and Izod impact test specimens shall be preconditioned for a minimum of 24 hours at 65 to 85 degrees F. In case of dispute, however, the conditioning or preconditioning environment and the test environment for basis weight, thickness and bending qualities shall be 73.4 degrees $\pm$ 3.6 degrees F and 50 $\pm$ 5 percent relative humidity.

4.5.2 Deflection temperature at 66 psi. Except as specified in 4.5.1 and this paragraph, the deflection temperature at a fiber stress of 66 psi shall be determined in accordance with ASTM D648. Test specimens shall be produced from the same material used in fulfilling the contract with molding conditions being those recommended by the manufacturer of the material. The molds used for producing the test specimens shall

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conform to the requirements of ASTM D647 for 1/4 by 1/2 by 5 inch specimens.

4.5.3 Notched Izod impact strength at 0 degrees F. With the exceptions specified in 4.5.1 and this paragraph, the impact resistance at 0 degrees F shall be determined in accordance with Method D of ASTM D256. Test specimens shall be produced from the same material used in fulfilling the contract with the molding conditions to be those recommended by the manufacturer of the material. The molds used in producing the test specimens shall conform to the requirements of ASTM D647 for 1/4 by 1/2 by 5 inch specimens.

4.5.4 Basis weight. With the exceptions specified in 4.5.1 and this paragraph, basis weight shall be determined in accordance with ASTM D646. A minimum of five test units from the same plastic board sheet with minimum dimensions of 10 inches by 10 inches shall be used for a basis weight determination.

4.5.5 Thickness. With the exceptions specified in 4.5.1 and this paragraph, thickness shall be determined in accordance with TAPPI T411. Five representative test specimens, with minimum dimensions of 3.00 inches inches by 5 inches, shall be cut from each sheet of plastic board. The thickness of each test specimen shall be measured at five non-overlapping points.

4.5.6 Bending qualities. Specimens, 12 inches by 12 inches, shall be scored parallel and perpendicular to the direction of the ribs or corrugating medium using scoring devices recommended for the board being tested. The test specimen shall be inspected for score line surface breaks or cracks. Cracks or breaks occurring at the intersection of score lines shall be disregarded. The specimen shall then be folded 180 degrees toward one facing and 180 degrees toward the other along the scoreline both parallel and perpendicular to the direction of the ribs or corrugating medium. The specimens shall be inspected for breaks or cracks after folding.

4.5.7 Flute T-peel strength.

4.5.7.1 Apparatus. A tension testing machine capable of applying a tensile load and having the following characteristics:

a. A loading range so that the maximum load on the specimen falls between 15 and 85 percent of the upper limit of the loading range.

b. A constant rate of movement between heads of 1 inch per minute.

c. The machine shall be equipped with suitable grips capable of clamping the specimens firmly without slippage during the test.

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d. The machine shall be autographic with a chart that can be read in terms of inches of separation as one coordinate and applied load as the other coordinate.

e. The applied load as measured and recorded shall be accurate within ± 1 percent.

4.5.7.2 Test specimens. Five test specimens shall be cut from each of two plastic board sheets so that no portion of the test specimens is within three inches of any of the sheet edges. The test specimens shall be cut to the width necessary to include exactly 5 flute peaks on each facing. The five flute peaks on each facing shall form the only bonds of the flutes to the facings. The test specimens shall be 12 inches long. Using a sharp knife or razor blade, separate the facings at one end to a length of 3 inches by cutting through the corrugations being careful not to damage the facings.

4.5.7.3 Test procedure. Clamp one of the facings, separated in the preparation of the test specimens, in the upper grip and the other facing in the lower grip of the tension testing machine aligning the specimen so that during the test, the load is applied along a line parallel to the specimen length. Apply the load at a constant rate of separation of grips of 1 inch per minute. During the test, make an autographic recording of load versus grip separation. The test shall be continued for a minimum of 6 inches of separation after the initial peak is reached (i.e., when the load recorded reflects the force required for separation) or until one of the facings ruptures. The recorded load, after the initial peak or when one of the facings ruptures, shall not be less than the value specified in 3.7.

4.5.8 Flatness test. Specimens 12 inches by 12 inches shall be cut from the middle of the sheets. Condition the specimens in accordance with 4.5.1. Place the specimens on a flat surface and measure the maximum vertical distance from the surface to the underside of the specimen (see 3.6).

4.6 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5 and the marking requirements of MIL-STD-129 (see 5.2.1, 5.2.2, and 6.2).

5. PACKAGING

5.1 Preservation. Not applicable.

5.2 Packing.

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5.2.1 Level A. Plastic board sheets of one type, basis weight and size not exceeding a weight of 1000 pounds or a stacked height of 24 inches shall be packed in boxes meeting level A requirements. The boxes shall be constructed with inside dimensions providing a snug fit for the stack of plastic board sheets.

5.2.2 Level C. The plastic board sheets of one type, basis weight and size shall be packed for delivery in a manner which will ensure arrival at destination in satisfactory condition and which will permit acceptance by carrier at the lowest applicable rate. Shipping containers shall conform to the requirements of the Uniform Freight Classification Rules or National Motor Freight Classification Rules as applicable.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Plastic board is primarily intended for use in the fabrication of reuseable boxes. In addition, plastic board may be used for those packaging applications requiring a container material whose properties are unaffected by exposure to water. Plastic board may also be used in the fabrication of pads, sleeves, box liners, partitions and cut shapes. Plastic board, like most other plastic materials, exhibits the tendency to accumulate and store large quantities of electric charges. It is therefore not recommended for packaging flammable and explosive materials or materials subject to damage from electrostatic charges unless the package or pack has been designed to prevent such hazards.

6.2 Ordering data. Purchasers should include the following information in procurement documents:

a. Title, number, and date of this specification.

b. Type (see 1.2 and 3.7).

c. Basis weight (see 3.8.1). It may be desirable to specify a basis weight range such as 160 to 170 pounds per 1000 square feet.

d. Length and width of sheet (see 3.11).

e. Quantity (number of sheets).

f. Specify the raw material (polyethylene, polypropylene, or copolymer) if a specific material is required (see 3.2).

g. Specify color (see 3.3).

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h. Specify if food grade board is required (see 3.2).

i. Specify whether first article inspection is waived see (4.3).

j. Specify packing level (see 5.2).

k. Specify special marking if required (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a first production item. The first article should consist of a minimum of two sheets. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Plastic corrugated
Box liners
Packaging
Pads
Partitions
Plastic Panel
Box Panel
Plastic board
Plastic corrugated board
Sleeves

6.5 Disposability. One or more of the following methods shall be used to accomplish disposal of plastic board: reuse, recycling, baling, sanitary landfill, composting, incineration, or pyrolysis.

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Review activities:

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User activities:

Army - SM
Air Force - 80

Preparing activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-83668A	Plastic Board (For Packaging Applications)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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